

Chapter 2 exercises — Recurrence Relations

74 exercises, extracted from the textbook PDF.

Caveats. (1) Statements are auto-extracted; `fi` / `fl` ligatures dropped by the PDF text layer were repaired by a cleanup pass (a rare odd gap may remain). (2) A $\triangle$ marks exercises whose statement includes a **displayed formula** that did not survive extraction — read those in the book at the cited page. The book PDF is always authoritative.

#	p.	Exercise
2.1	46	Write recursive and nonrecursive programs to compute values for the Fibonacci recurrence and try to use each to compute F_{20} . Explain the behavior of each program in this case.
2.2	46	How many arithmetic operations are used by Program 2.1, as a function of $N_{\max}$?
2.3	46	Write a recursive program to compute values using the recurrence (1) directly. How does the number of arithmetic operations used by this program compare with Program 2.1 (see the previous exercise)?
2.4	46	Estimate how many operations would be required by both recursive and nonrecursive programs to compute values using the recurrences (2) and (3).
2.5	46	Write a program to compare quicksort, its median-of-three variant, and radix-exchange sort, calculating values from the recurrences given in Chapter 1. For quicksort, check values against the known solution; for the others, make conjectures about properties of the solution.
2.6	47	Solve the recurrence $a_n = a_{n-1} + a_{n-2}$
2.7	47	Solve the inhomogeneous recurrence $a_n = a_{n-1} + a_{n-2} + r$
2.8	47	For f linear, express the solution to the recurrence $a_n = f(a_{n-1}, a_{n-2})$
2.9	49	Solve the recurrence $a_n =$
2.10	49	Solve the recurrence $a_n = a_{n-1} + (-1)^n n$
2.11	49	Solve the recurrence $na_n = (n-2)a_{n-1} + 2$
2.12	49	Solve the recurrence $a_n = 2a_{n-1} + 1$ (Hint : Divide both sides by 2^n .)
2.13	51	Solve the recurrence $a_n =$
2.14	51	Write down the solution to for $n > t$
2.15	51	Solve the recurrence $na_n = (n+1)a_{n-1} + 2n$
2.16	51	Solve the recurrence $na_n = (n-4)a_{n-1} + 12nH_n$
2.17	51	[Yao] ("Fringe analysis of 2-3 trees [25]") Solve the recurrence $AN = AN_{-1} -$
2.18	52	Define $b_n = a_n - \alpha$ with a_n and α defined as above. Find an approximate formula for b_n when n is large and a_0 is between 0 and 1.

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2.19	52	Show that $a_n = \cos(a_{n-1})$ converges when a_0 is between 0 and 1, and compute $\lim_{n \rightarrow \infty} a_n$ to ve decimal places.
2.21	54	Prove that $a_n = \Theta(1/n)$. Compute initial terms and try to guess a constant c such that a_n is approximated by c/n . en find a rigorous proof that na_n tends to a constant.
2.22	54	[De Bruijn] Show that the solution to the recurrence $a_n = \sin(a_{n-1})$
2.23	54	What happens when $f'(\alpha) > 1$?
2.24	54	State sufficient criteria corresponding to the three cases above for local convergence (when a_0 is sufficiently close to α) and quantify the speed of convergence in terms of $f'(\alpha)$ and $f''(\alpha)$.
2.25	55	Give a recurrence that has the solution $a_n = 4^n - 3^n + 2^n$.
2.26	58	Explain how to solve an inhomogeneous recurrence of the form $a_n = x_1 a_{n-1} + x_2 a_{n-2} + \dots + x_t a_{n-t} + r$
2.27	58	Give initial conditions a_0, a_1 for which the solution to $a_n = 5a_{n-1} - 6a_{n-2}$
2.28	59	Give initial conditions a_0, a_1 , and a_2 for which the growth rate of the solution to $a_n = 2a_{n-1} - a_{n-2} + 2a_{n-3}$ for $n > 2$ is (i) constant, (ii) exponential, and (iii) fluctuating in sign.
2.29	59	Solve the recurrence for $n > 1$ with $a_1 = 2$ and $a_0 = 1$.
2.30	59	Solve the recurrence $a_n = 2a_{n-1} - a_{n-2}$
2.31	59	Solve the recurrence $a_n = a_{n-1} - a_{n-2}$
2.32	59	Solve the recurrence $2a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3}$
2.33	59	Find a recurrence describing a sequence for which the order of growth decreases exponentially for odd-numbered terms, but increases exponentially for even-numbered terms.
2.34	59	Give an approximate solution for the “third-order” Fibonacci recurrence (3)
2.35	60	Solve the recurrence $n(n-1)a_n = (n-1)a_{n-1} + a_{n-2}$
2.36	60	Give a simple algorithm to determine whether a given monomial $s_1^{i_1} s_2^{i_2} \dots s_p^{i_p} t_1^{j_1} t_2^{j_2} \dots t_q^{j_q}$ appears in the expansion of a_n . How many such monomials are there?
2.37	62	Give exact formulae for b_n and a_n .

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2.38	62	Solve the recurrence $\sqrt{a_n} = 1 + a_{2n-1}$
2.39	62	From the above discussion, solve the register allocation recurrence for $a_0 = 3, 4$. Discuss what happens for $a_0 = 3/2$.
2.40	63	Solve the register allocation recurrence for $a_0 = 2 + \epsilon$, where ϵ is an arbitrary fixed positive constant. Give an accurate approximate answer.
2.41	63	Find all values of the parameters α, β , and γ such that $a_n = \alpha a_{2n-1} + \beta a_{n-1} + \gamma$ reduces to $b_n = b_{2n-1} - 2$ by a linear transformation ($b_n = f(\alpha, \beta, \gamma)a_n + g(\alpha, \beta, \gamma)$). In particular, show that $a_n = a_{2n-1} + 1$ does not reduce to this form.
2.42	63	[Melzak] Solve the recurrence $a_n = 2a_{n-1}$
2.43	64	Solve the recurrence $a_n =$
2.44	64	Consider the recurrence $a_n = 1/(s_n + t_n a_{n-1})$
2.45	66	Solve the quicksort recurrence for $f(n) = n^3$.
2.46	66	[Greene and Knuth] Solve the quicksort median-of-three recurrence (see equation (4) in Chapter 1) using the repertoire method. (See [18] or [24] for a direct solution to this recurrence using differencing and summation factors, and see Chapter 3 for a solution using generating functions.)
2.47	67	Solve the recurrence $a_n = 2/(n + a_{n-1})$
2.48	67	Use bootstrapping to show that the number of compares used by median-of-three quicksort is $\alpha N \ln N + O(N)$. Then determine the value of α .
2.49	67	[Greene and Knuth] Use bootstrapping to show that the solution to $a_n =$
2.50	69	Find the asymptotic growth of the solution to the “perturbed” Fibonacci recurrence $a_{n+1} = 1 + a_n + 1 - a_{n-1}$ for $n > 1$ with $a_0 = 0$ and $a_1 = 1$.
2.51	69	Solve the recurrence $a_n = na_{n-1} + n^2 a_{n-2}$
2.52	69	[Aho and Sloane]
2.53	69	Solve the following perturbation of the Fibonacci recurrence: $a_n = 1 - (a_{n-1} + a_{n-2})$ for $n > 1$ with $a_0 = a_1 = 1$. Try a solution of the form $n^\alpha \phi^n$ and identify α .
2.54	73	What is the number of comparisons used during an unsuccessful search with binary search in a table of size N in the best case?

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2.55	73	Consider a “ternary search” algorithm, where the file is divided into thirds, two comparisons are used to determine where the key could be, and the algorithm is applied recursively. Characterize the number of comparisons used by that algorithm in the worst case, and compare it to a binary search.
2.56	75	By considering the rightmost bits, give a direct proof that the number of bits in the binary representations of all the numbers less than N satisfies (4), but with an additive term of $N - 1$ instead of N .
2.57	75	Prove that $N \lfloor \lg N \rfloor + 2N - 2 \lfloor \lg N \rfloor + 1 = N \lceil \lg N \rceil + N - 2 \lceil \lg N \rceil$ for all positive N . (See Exercise 1.4.)
2.58	79	Give recurrences for the functions plotted in Figure 2.5.
2.59	79	Derive recurrences for R_N similar to those given above for P_N .
2.60	79	Plot the solution to the recurrence $A_N = A_{\lfloor N/2 \rfloor} + A_{\lfloor N/2 \rfloor} + \lfloor \lg N \rfloor$
2.61	79	Plot the solution to the recurrence $B_N = 3B_{\lfloor N/2 \rfloor} + N$
2.62	79	Plot the solution to $D_N = D_{\lfloor N/2 \rfloor} + D_{\lfloor N/2 \rfloor} + CN$
2.63	80	Take the binary representation of N , reverse it, and interpret the result as an integer, $\rho(N)$. Show that $\rho(N)$ satisfies a divide-and-conquer recurrence. Plot its values for $1 \leq N \leq 512$ and explain what you see.
2.64	80	What is the average length of the initial string of 1s in the binary representation of a number less than N , assuming all such numbers are equally likely?
2.65	80	What is the average length of the initial string of 1s in a random bitstring of length N , assuming all such strings are equally likely?
2.66	80	What is the average and the variance of the length of the initial string of 1s in a (potentially infinite) sequence of random bits?
2.67	80	What is the total number of carries made when a binary counter increments N times, from 0 to N ?
2.68	84	Give explicit formulae for c_1 and c_2 . Start by doing the case $\delta = 0$.
2.69	85	Plot the periodic part of the solution to the recurrence $a_N = 3a_{\lfloor N/3 \rfloor} + N$
2.70	85	Answer the previous question for the other possible ways of dividing a problem of size N into three parts with the size of each part either $\lfloor N/3 \rfloor$ or $\lceil N/3 \rceil$.
2.71	85	Give an asymptotic solution to the recurrence $a(x) = \alpha x / \beta + 2x$

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2.72	85	Give an asymptotic solution to the recurrence for $N > 2$ with $a_1 = a_2 = a_3 = 1$.
2.73	85	Give an asymptotic solution to the recurrence $a_N = a_{N/2} + a_{N/4} + N$
2.74	85	Consider the recurrence $a_n = a_{f(n)} + \text{flag}(n) + a_{h(n)} + 1$
2.75	85	Consider the recurrence $a_n = a_{f(n)} + \text{flag}(n) + 1$